Lab 5

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In [ ]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Read the image
image = cv2.imread('c1.jpg')
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB) # Convert from BGR to RGB

# Gaussian sigma
sigma=2

#parameter
amount=1.5

# Convert image to float32 for processing
image_float = image.astype(np.float32) / 255.0

# Apply Gaussian blur
blurred = cv2.GaussianBlur(image_float, (0, 0), sigmaX=sigma,
sigmaY=sigma)

# Perform Unsharp Masking
sharpened = np.clip(image_float + amount * (image_float - blurred), 0,
1)

# Plot original, blurred, and sharpened images
fig, ax = plt.subplots(1, 3, figsize=(15, 5))
ax[0].imshow(image)
ax[0].set_title("Original Image")
ax[0].axis("off")
ax[1].imshow(blurred)
ax[1].set_title("Blurred Image")
ax[1].axis("off")
ax[2].imshow(sharpened)
ax[2].set_title("Sharpened (Unsharp Masking)")
ax[2].axis("off")
plt.show()
```



In [5]:
```
#Task 1

filter = np.ones((7,7), np.float32) / 49

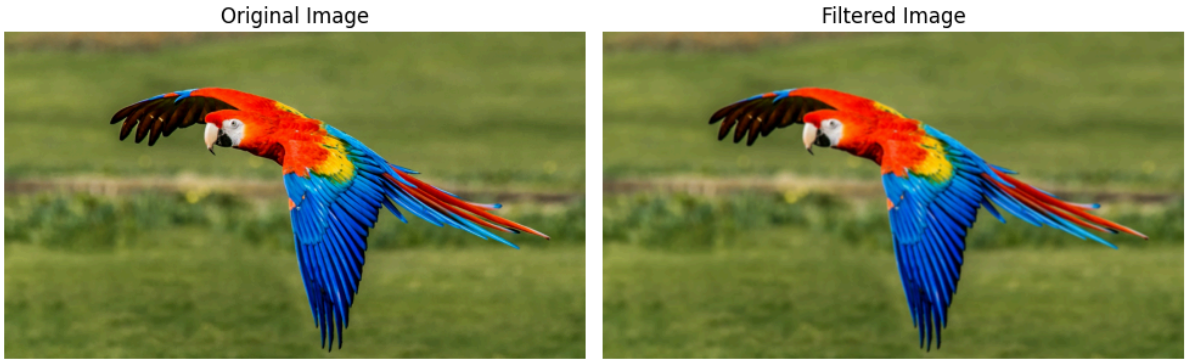
filtered_image = cv2.blur(image_float, (7, 7))

plt.figure(figsize=(10, 5))

plt.subplot(1, 2, 1)
plt.imshow(image_float)
plt.title('Original Image')
plt.axis('off')

plt.subplot(1, 2, 2)
plt.imshow(filtered_image)
plt.title('Filtered Image')
plt.axis('off')

plt.tight_layout()
plt.show()
```



In [6]:
```
#Task 2

gaussian_5 = cv2.GaussianBlur(image, (5, 5), 0)

sharpened = np.clip(image_float - gaussian_5, 0, 1)

gaussian_21 = cv2.GaussianBlur(image, (21, 21), 0)

plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1)
plt.imshow(image)
plt.title('Original Image')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(gaussian_5)
plt.title('Gaussian Filter (5x5)')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(gaussian_21)
plt.title('Gaussian Filter (21x21)')
plt.axis('off')

plt.tight_layout()
plt.show()
```



In [7]:
```
#Task 3

laplacian = cv2.Laplacian(image_float, cv2.CV_32F)

sharpened = np.clip(image_float - laplacian, 0, 1)

plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1)
plt.imshow(image_float)
plt.title('Original Image (Float32)')
plt.axis('off')

plt.subplot(1, 3, 2)

plt.imshow(np.clip(laplacian, 0, 1))
plt.title('Laplacian Filter Result')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(sharpened)
plt.title('Sharpened Image')
plt.axis('off')

plt.tight_layout()
plt.show()
```

